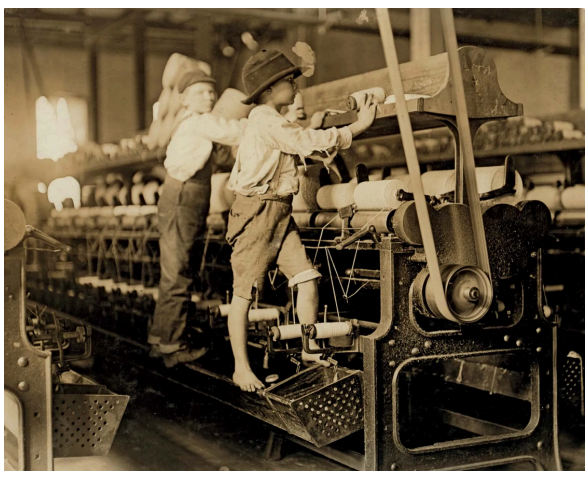
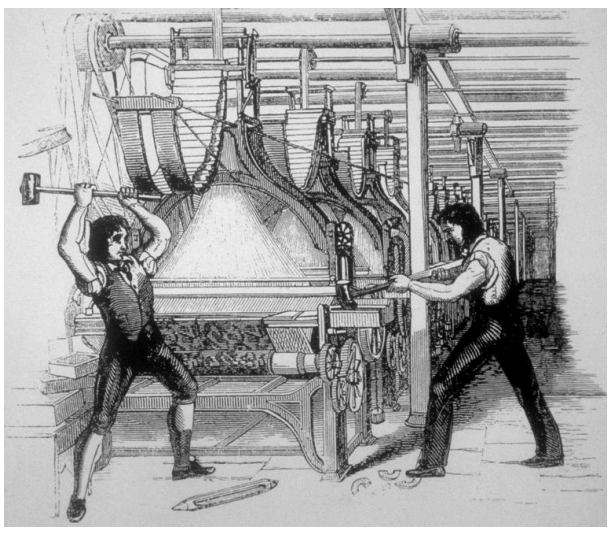
# Automation

## The Industrial Revolution, Technological Unemployment, and Luddites

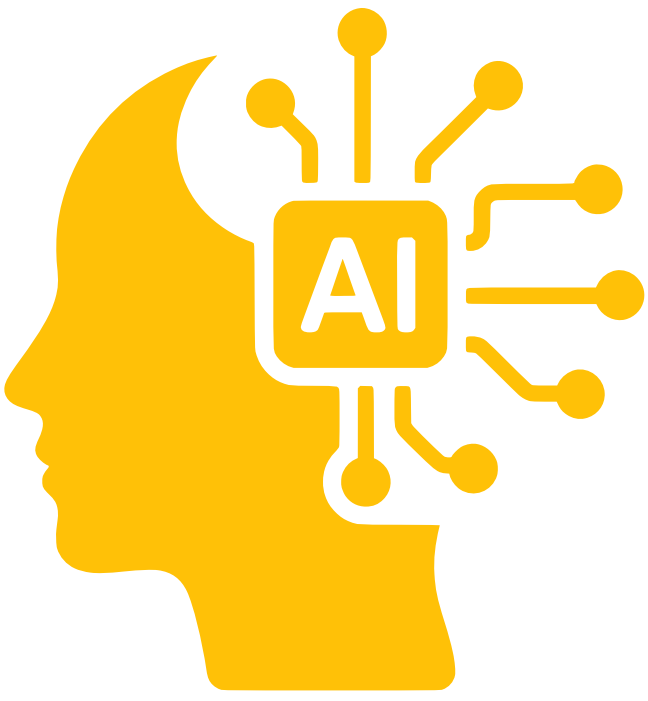


Technological unemployment is defined as the loss of jobs caused by new tools, automation and artificial intelligence. One of the major cases of mass technological unemployment is during the British Industrial revolution. The industrial revolution started the development of automated machines. These machines resulted in job losses in many industries. One of the most impacted areas was the textile industry, where the mechanisation of looms became the symbol of automation and job loss.

Following the massive layoffs, a new movement arose around the textile industry that aimed to protest against the mechanisation of looms, poor working conditions and the mass layoffs called the Luddites movement. The label Luddite has evolved to indicate anyone who is against new technology or development in the industry.



## The AI Revolution and the Neo-Luddites




The AI revolution is frequently compared to the industrial revolution, as AI technology shows a potential in replacing and automating jobs, causing major layoffs just like its industrial counterpart. However, the major difference between the AI revolution and the industrial revolution is that they will replace different jobs. Artificial Intelligence is moving to replace white collar jobs, or jobs that were considered specialized in the intellectual requirements and capabilities of the human laborer. This means that jobs that were previously considered safe from the risk of technological development may become the most vulnerable to replacement by AI.

The AI revolution, just like the industrial revolution, has birthed a movement that opposes technological development and automation. With the suffix “neo” meaning “new”, the Neo-Luddite movement is a group of philosophers that oppose the rapid technological development of modern society, especially regarding AI and automation. Luddite movements, however, do not simply reject progress and new technology, they protest against the ways in which the technology is used against them.



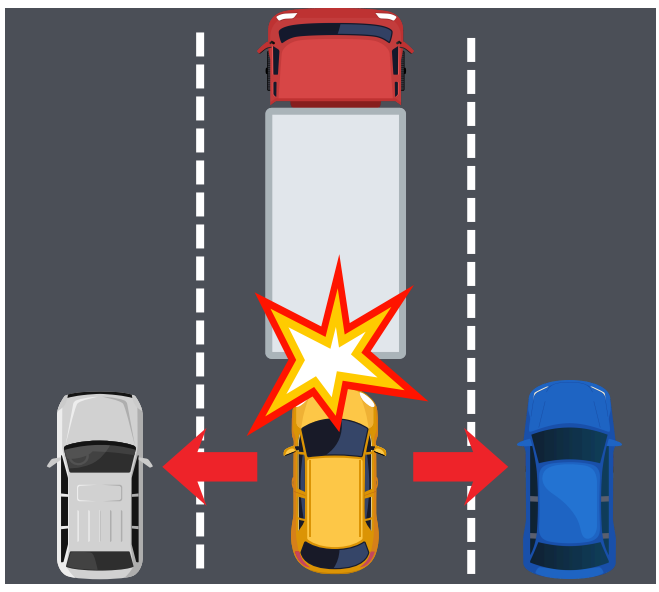
# Automation

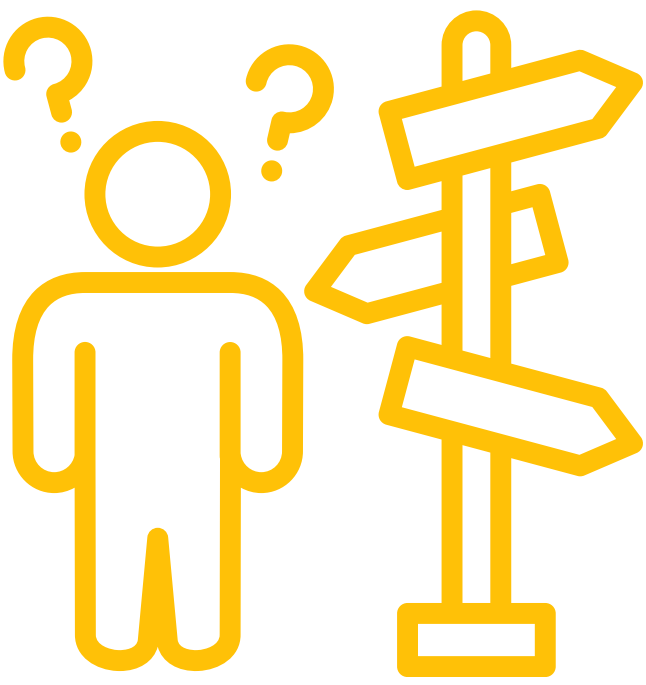
## AI Automation and Accountability



The key issue with AI and automation is the accountability aspect. Without humans that make decisions, there is no one to take responsibility for the actions that an AI model takes. A thought experiment illustrates this.

Assume that you are driving down the road, and the truck in front of you releases what they had been transporting. If you do not swerve, it is certain death. There are two cars either side of you. Which way do you swerve? What if one of the cars were replaced by a motor cycle? What if the person in the car is pregnant? What if the car has children? What if the biker isn't wearing a helmet? Do you swerve at all?





A human can make the excuse that they couldn't think properly in that moment and that they moved instinctually. Now imagine that this is a self driving car. AIs are capable of making thousands of decisions in a second, and they don't panic. This means that anything that the AI does becomes explainable, and blame-able. The AI makes a choice, a choice that harms someone inevitably. Should AIs be allowed to make these critical decisions that may put someone above others? Who takes accountability for the decisions that the AI makes in that moment?

To extend this to a daily situation, imagine that there is a car crash between two self driving cars. Traditionally, there would be arguments between the drivers on who's fault it is, and who pays who for what kind of damage. However, because the car crash was the result of an AI decision, not a human one, the responsibility can not go anywhere. Who is to blame for the car crash? The AI model's developers? The company that made the cars? The area of accountability becomes increasingly vague as decision making is yielded over to AI models.





Raine v. OpenAI is a case that represents this accountability issue. Raine v. OpenAI was filed by Matthew and Maria Raine claiming the wrongful death of their 16 year old son who was apparently given suicide coaching by OpenAI's AI chatbot, ChatGPT. The case is ongoing.